

Indian Statistical Institute
First Semester Examinations: 2025-26

Course Name: M. Math, 2nd year
Subject Name : Topology III
Maximum Marks: 20, Duration: 1.5 hours
Date: 06.03.2025, 11:00 AM - 12:30 PM

- Answer as many questions as you can.
- Maximum marks is 20.
- You may use any results proved in class. Any other results (including those in homework problem sets) require proof.

✓ Suppose that X, Y, Z are pointed spaces, and define the space $F_2(X, Y, Z) \subset X \times Y \times Z$ as

$$F_2(X, Y, Z) = \{(x, y, z) \in X \times Y \times Z \mid \text{at least one of } x, y, z \text{ is } *\}.$$

Prove that the inclusion $F_2(X, Y, Z) \rightarrow X \times Y \times Z$ is surjective on homotopy groups.

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- ✓ 2. Let Σ_g denote the surface of genus g . Show that there is a degree 1 map from $\Sigma_g \rightarrow \Sigma_h$ if and only if $g \geq h$. 6
3. Let $f : X \rightarrow Y$ be a based map of path connected CW complexes. Show that there is a CW complex Z_n and maps $i_n : X \rightarrow Z_n, j_n : Z_n \rightarrow Y$ with $f = j_n \circ i_n$, such that i_n is an isomorphism on π_k for $k \leq n$ and j_n is an isomorphism on π_k for $k \geq n + 1$. 10